Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 119405 | *Subject: Structural Analysis* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In the force method, the unknowns are primarily
(i) Redundant forces
(ii) Rotations only
(iii) Member stiffness
(iv) Joint displacements

(b) Flexibility coefficient represents
(i) Moment caused by unit load
(ii) Displacement caused by unit force
(iii) Stress per unit strain
(iv) Force required for unit displacement

(c) Castigliano's theorem is commonly used in force method to compute
(i) Redundant forces
(ii) Strains
(iii) Displacements
(iv) Reactions

(d) The point of contraflexure is the point where
(i) Shear force is zero
(ii) Bending moment is maximum
(iii) Bending moment changes sign
(iv) Deflection is maximum

(e) A Castigliano's Second Theorem is applicable when
(i) Material is nonlinear
(ii) Structure behaves elastically
(iii) Loads are dynamic
(iv) Temperature effects dominate.

(f) In moment distribution method, the distribution factor at a joint depends on
(i) Load on the member
(ii) Flexural rigidity of connected members
(iii) Length of member
(iv) Support conditions only

(g) For a linearly elastic structure, the strain energy is equal to
(i) Total potential energy
(ii) Plastic work
(iii) Kinetic energy
(iv) External work done

(h) The carry-over factor in moment distribution method for a prismatic member is
(i) 1
(ii) 1/2
(iii) 1/3
(iv) 2/3

(i) Cantilever method is best suited for analysis of
(i) Single-storey frames
(ii) Low-rise frames
(iii) Tall multi-storey frames
(iv) Simply supported beams

(j) The portal method generally neglects the effect of
(i) Axial forces in columns
(ii) Bending moments in beams
(iii) Shear forces
(iv) Storey shear distribution

Q.2 (a) State and explain the Principle of Minimum Potential Energy. **[7]**
(b) State Castigliano's second theorem and explain how it differs from the first theorem. **[7]**

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Q.3 (a) A cantilever beam with support at left end have span length of L is subjected to a tip moment of M. Evaluate slope and deflection at a point (3L/4) from left support. Assume EI of the given beam to be constant. [7]

(b) Name the type of rolling loads for which the absolute maximum bending moment occurs at the mid span of a beam? [7]

Q.4 (a) State Muller Breslau's principle [4]

(b) A simply supported beam has a span of 16 m is subjected to a UDL (dead load) of 5 kN/m and a UDL (live load) of 8 kN/m (longer than the span) traveling from left to right. Draw the ILD for shear force and bending moment at a section 4 m from the left end. Use these diagrams to determine the maximum shear force and bending moment at this section. [10]

Q.5 (a) A cantilever beam of length L carries a point load P at the free end and a moment M at the same point. Using the moment area method, determine the free-end slope and deflection. [7]

(b) Discuss about the sway in frames also discuss about sway correction. [7]

Q.6 Analyze the continuous beam ABC shown in Fig. 6.1 by slope deflection method and sketch the bending moment and shear force diagram. Take EI = constant. [14]

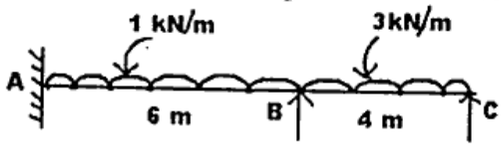
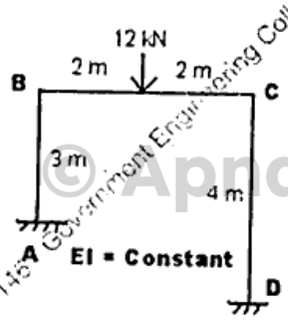


Fig.

Q.7 Analyse the portal frame shown in figure 7.1 by slope deflection method and determine the shear force and bending moment diagram. [14]



Q.8 For the continuous beam shown in Fig. 7.1, Determine the support reactions and draw the shear force and bending moment diagram using Moment Distribution method. [14]

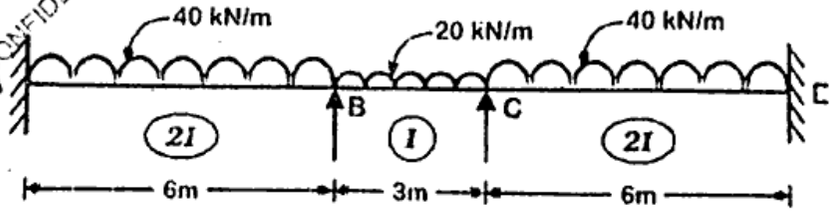


Fig. 3.9(a)

Q.9 (a) Compare force methods, displacement methods, and energy methods of structural analysis. [7]

(b) Discuss about the Portal and cantilever method for the analysis of multi-storey frames. [7]

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